



## 1 Background and Motivation

A **conventional transceiver design** treats coding, modulation, channel estimation and detection as separate blocks. Each block is optimal for its own task, but not for the system as a whole. Consequently, the end-to-end performance is limited. To address this practical issue, the signal processing blocks at the transmitter and the receiver can be replaced by deep neural networks which are jointly optimised as an **autoencoder** [1].

### Research questions:

- How much performance gain does end-to-end learning offer over a **well-optimised classical receiver**?
- Can the end-to-end system be trained **without a differentiable channel model**?

## 2 Objectives and Contributions

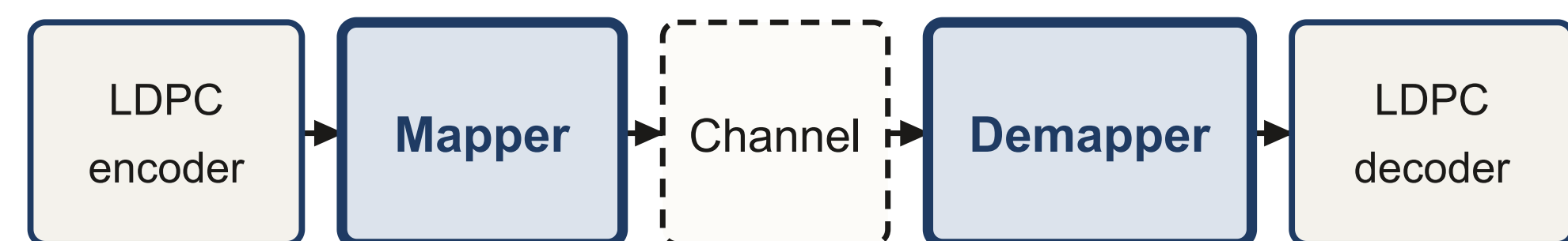
### Objectives

- Compare an end-to-end learned transceiver with a **matched** classical system over AWGN and Rayleigh block-fading channels with **pilot-estimated CSI**.
- Evaluate transmitter training by reinforcement learning when **no differentiable channel model** is available.

### Contributions

- Develop a fair evaluation that includes **pilot overhead** and **channel-estimation error**.
- Show that reinforcement learning can train the transmitter with only a small performance penalty compared with gradient-based training.
- Show that the proposed **end-to-end communication system** outperforms a conventional channel estimation method.

## 3 System Model



|                  |                         |                                   |
|------------------|-------------------------|-----------------------------------|
| <b>Learned</b>   | trainable constellation | neural network                    |
| <b>Classical</b> | 64-QAM, Gray            | a posteriori probability demapper |

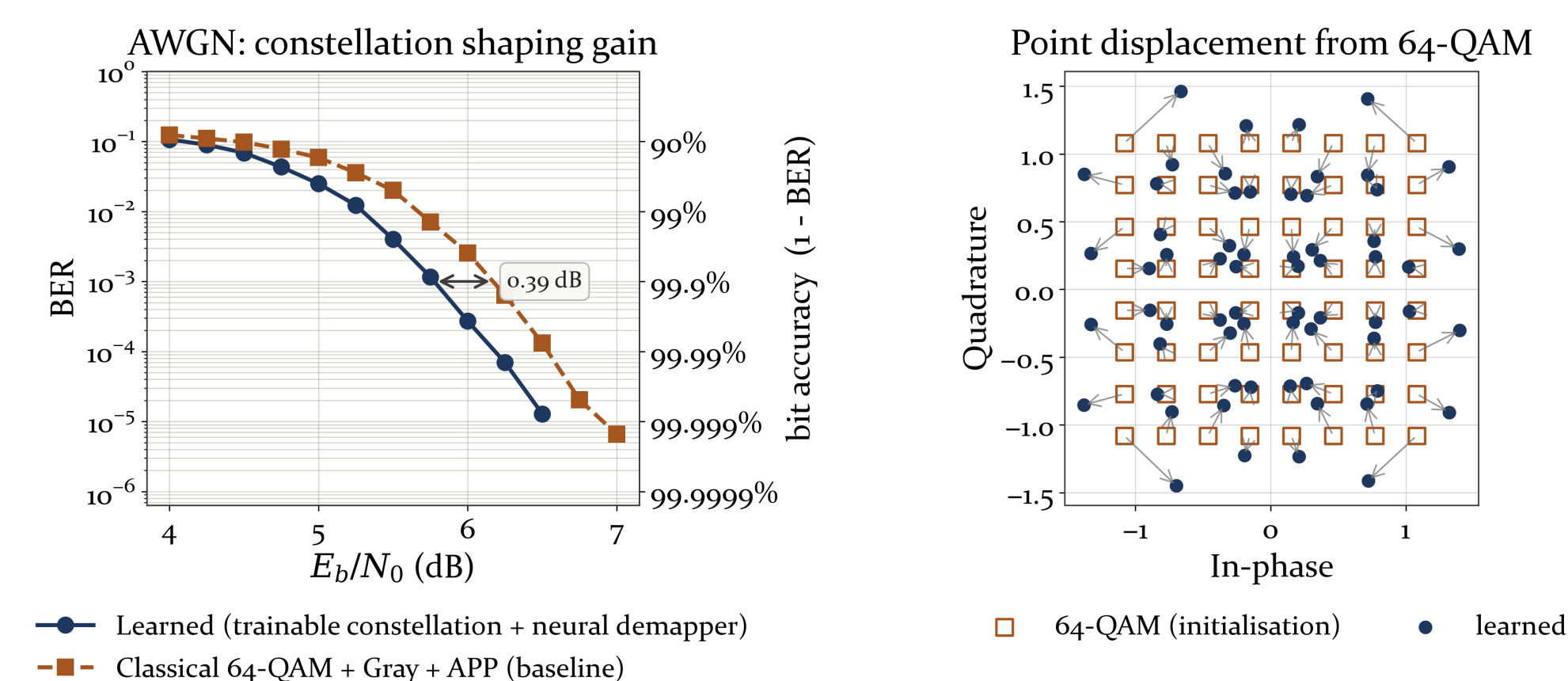
Channel: AWGN or Rayleigh block fading,  $L = 25$ ,  $P$  pilot symbols, LS estimate

The two systems differ only in the shaded blocks. They share the rate-1/2 5G LDPC code, the channel model and the random seeds, so any difference in bit error rate comes from the mapper and the demapper.

Let  $R$  be the code rate,  $L$  the number of symbols in one fading block,  $P$  the number of pilot symbols in that block,  $N_0$  the noise power spectral density and  $\hat{h}$  the channel coefficient estimated from the pilots. Both systems are evaluated on the same terms. The effective code rate is  $R(L-P)/L$ , and after zero-forcing the effective noise is  $N_0(1+1/P)/|\hat{h}|^2$ .

Seeds are fixed, and every number is measured in the accompanying notebook. Built with Sionna [5] and PyTorch. **Scope:** flat block fading only, with no multipath, Doppler or OFDM.

## 4 Where the AWGN Gain Comes From



Under AWGN, a posteriori probability (APP) demapping is already optimal, so the improvement cannot come from the receiver. It comes from the transmitter. The learned constellation moves away from the regular 64-QAM lattice and gives more power to the symbols that are most often confused.

## References

- [1] T. O'Shea and J. Hoydis, "An Introduction to Deep Learning for the Physical Layer," arXiv:1702.00832, 2017.
- [2] H. Ye, G. Y. Li and B.-H. F. Juang, "Power of Deep Learning for Channel Estimation and Signal Detection in OFDM," arXiv:1708.08514, 2017.
- [3] S. Smeka J. et al., "Accelerating Meta-Learning with the Enhanced Reptile Algorithm for Rapid Adaptation in Neural Networks," SCSE, IEEE, 2025.
- [4] N. C. Luong et al., "Applications of Deep Reinforcement Learning in Communications and Networking: A Survey," IEEE Commun. Surveys Tuts., vol. 21, no. 4, 2019.
- [5] NVIDIA Sionna, autoencoder tutorial notebook, which this work extends.

## 5 Performance Analysis

**Table 1** —  $E_b/N_0$  needed for a given bit error rate (dB, lower is better). 64,000 codewords per point. Each estimate is based on at least 30 observed block errors.

| System                                  | 1e-2        | 1e-3        | 1e-4        |
|-----------------------------------------|-------------|-------------|-------------|
| AWGN, classical 64-QAM + APP            | 5.67        | 6.17        | 6.54        |
| <b>AWGN, learned (channel gradient)</b> | <b>5.30</b> | <b>5.78</b> | <b>6.18</b> |
| AWGN, learned (RL, no channel model)    | 5.42        | 5.92        | 6.31        |
| Rayleigh, classical 64-QAM + APP        | 11.26       | 13.30       | 14.79       |
| Rayleigh, learned                       | 11.25       | 13.19       | 14.94       |

- Rayleigh fading requires **5.9 to 8.8 dB** more than AWGN, while the choice of receiver changes the result by less than **0.4 dB**.
- On AWGN the learned system gains **0.37, 0.39 and 0.36 dB** at BER 1e-2, 1e-3 and 1e-4. APP demapping is already optimal here, so the gain comes from **constellation shaping**.
- With only two pilot symbols per block, leaving the channel-estimation term out of the effective noise understates it by **1.76 dB**.
- Under Rayleigh with estimated CSI there is a **small gain** at 1e-2 and 1e-3, of 0.01 and 0.11 dB, and a loss of 0.15 dB at 1e-4. The value at 1e-4 changed sign between two runs.

## 6 Why AWGN-Trained Weights Fail on Fading

Reused on a fading channel without retraining, the AWGN weights give a BER that increases with SNR. A real system should not behave this way, so we kept every weight fixed and changed one thing at a time:

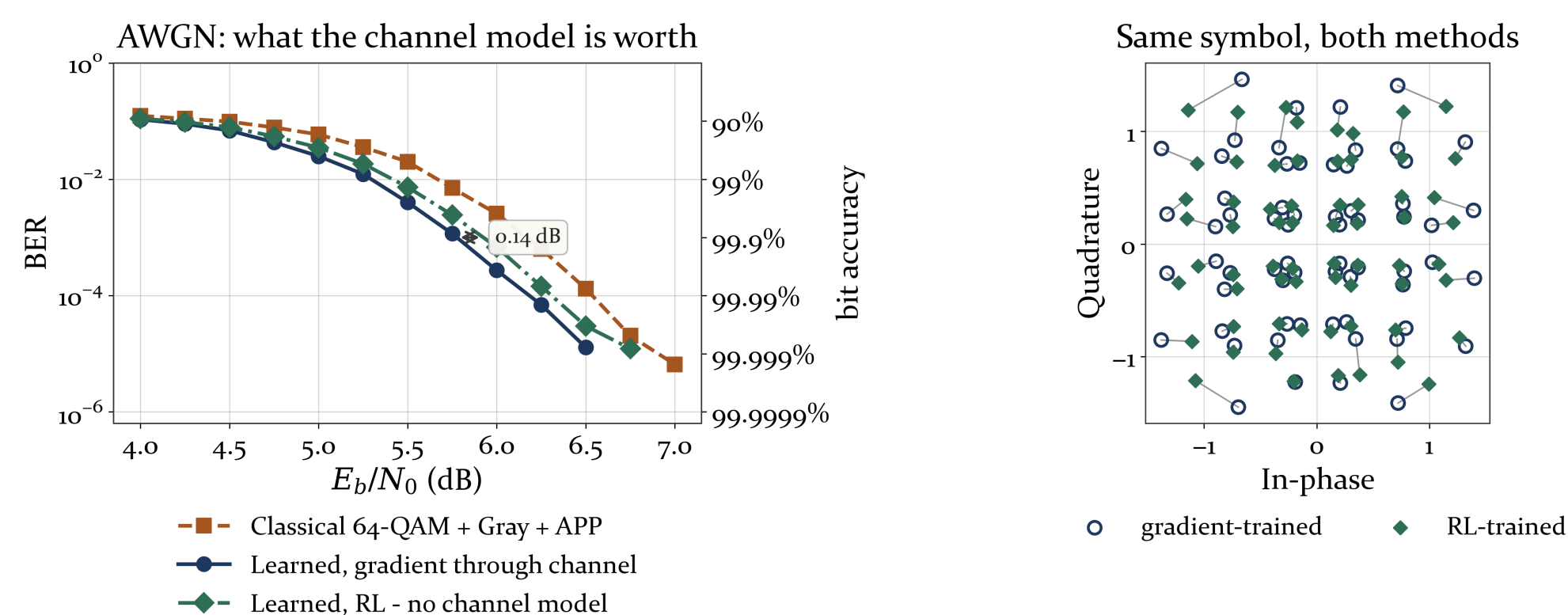
- Clamping the network outputs changed the BER by less than 4%, and the curve still increased.
- Removing the symbols in deep fades made the BER worse above 10 dB.
- Clamping the noise value passed to the demapper into the range it saw during training **restored a decreasing curve**, without changing any weight.

The cause is on the input side. At 18 dB, **89%** of symbols have an effective noise below anything seen in training, against 4.6% in deep fades. Retraining gives BER **1e-3** at 13.2 dB and **1e-4** at 14.9 dB.

## 7 Training Without a Channel Model

All the results above used backpropagation through a differentiable channel model. This may not be feasible in practice, where the channel acts as a black box and only its inputs and outputs can be observed. There is then **no channel layer** connecting the two neural networks at the transmitter and the receiver, because the channel is not differentiable.

Instead we feed the training loss from the receiver back to the transmitter, which is then trained by **reinforcement learning** [4]. The two networks are still jointly optimised, without requiring a channel model.



- Removing the channel model costs **0.12, 0.14 and 0.13 dB** at BER 1e-2, 1e-3 and 1e-4.
- The reinforcement-learned system is still **0.23 to 0.25 dB** better than classical 64-QAM.
- The two training methods place each symbol at almost the same position, so reinforcement learning converges to a similar constellation.

## 8 Conclusion

On AWGN, end-to-end learning gives a gain of **0.37 dB**. Under Rayleigh fading with estimated CSI the gain is small: 0.01 and 0.11 dB at 1e-2 and 1e-3, and -0.15 dB at 1e-4. The end-to-end system can also be trained **with no channel model**, at a cost of **0.12 to 0.14 dB**.

## 9 Next Steps

- Deploy on real hardware**, such as a software-defined radio, and measure whether the gain still appears outside simulation.
- Meta-learning for fast adaptation** [3]. The section above shows that the receiver fails when its input distribution changes.
- Extend to OFDM and multipath** [2], where a convolutional demapper can use the correlation between neighbouring subcarriers.
- Replace zero-forcing with MMSE equalisation**, which keeps the effective noise bounded.
- Better performance under fading**. We will continue to develop the system on Rayleigh and other realistic channel models.